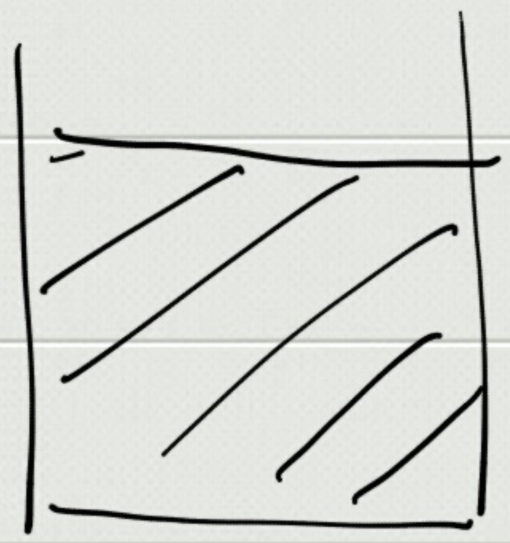
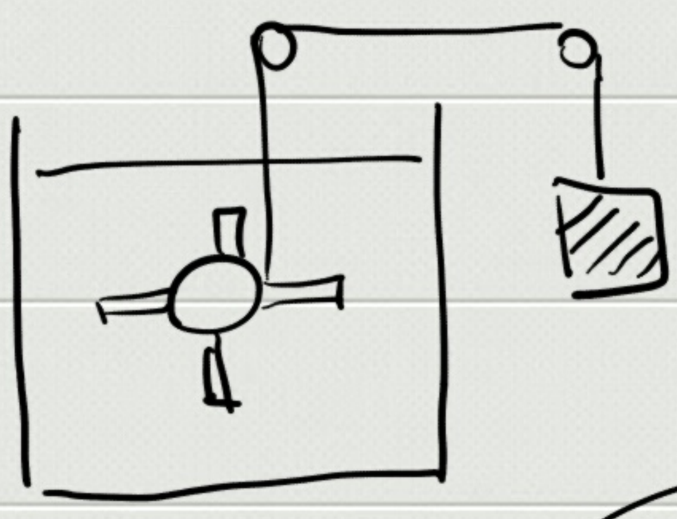
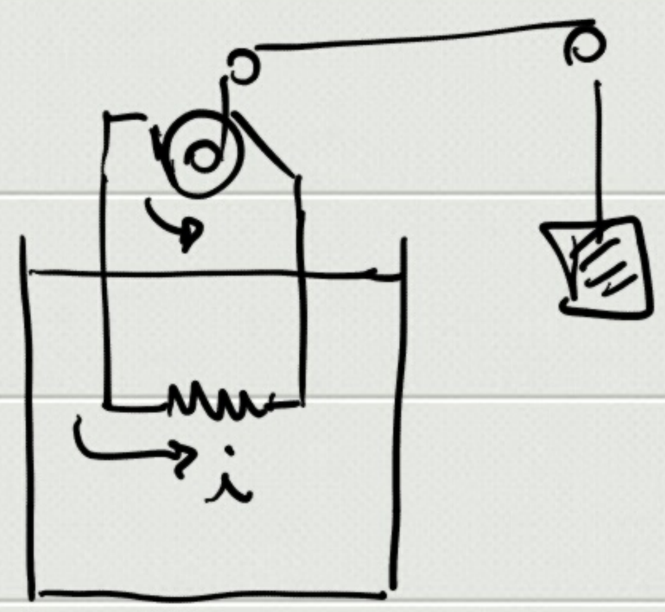
$i \to f$ $\qquad W_{i\to f} = \int_i^f p(V,T)\, dV \quad (\Rightarrow \delta W)$

(trasf. quasi-statica)



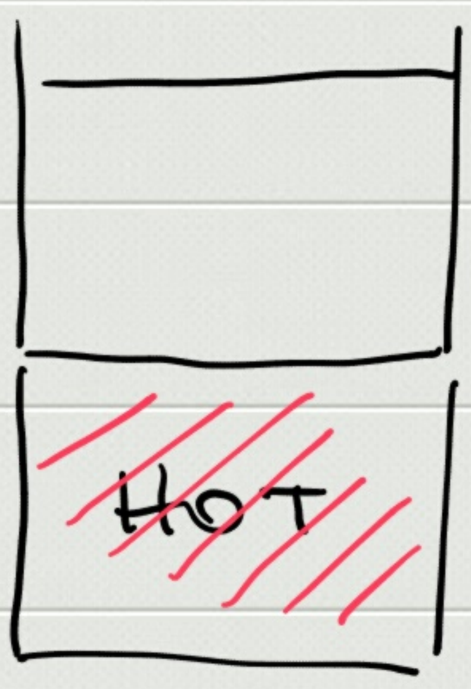
$T_i \to T_f$

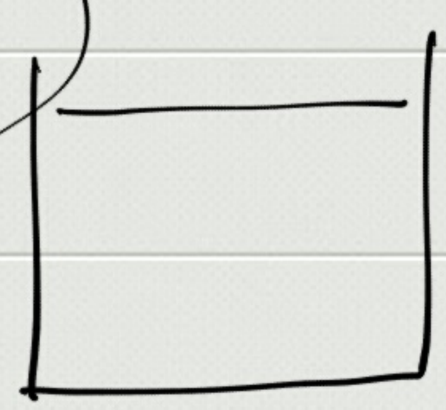




lavoro

$T_i \to T_f$



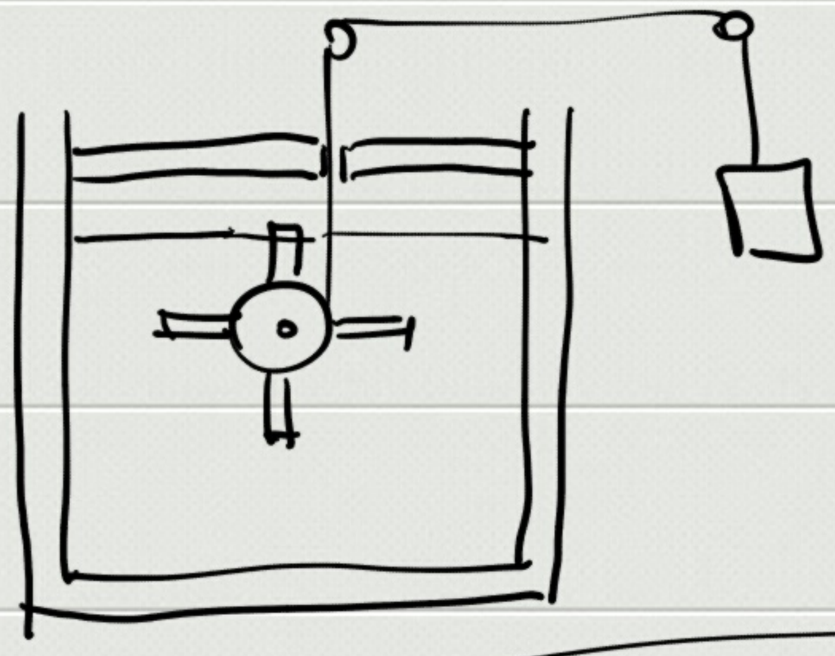


puramente termici

(no lavoro)

"~~Calorico~~" $\Rightarrow$ $\boxed{\text{Calore}}$

Joule



esp. adiabatico

$T_i \to T_f$

$$\boxed{W_{adiab} = \text{costante}}$$

forze cons.

$$W_{A\to B} = \int_A^B \bar{F}\, d\bar{s} = -(E_{p,B} - E_{pA}) = -\Delta E_p$$

$$\boxed{W_{adiab,\, i\to f} = -\Delta U} = -(U_f - U_i)$$

$$\boxed{U = \text{energia interna}}$$

Processi puramente termici

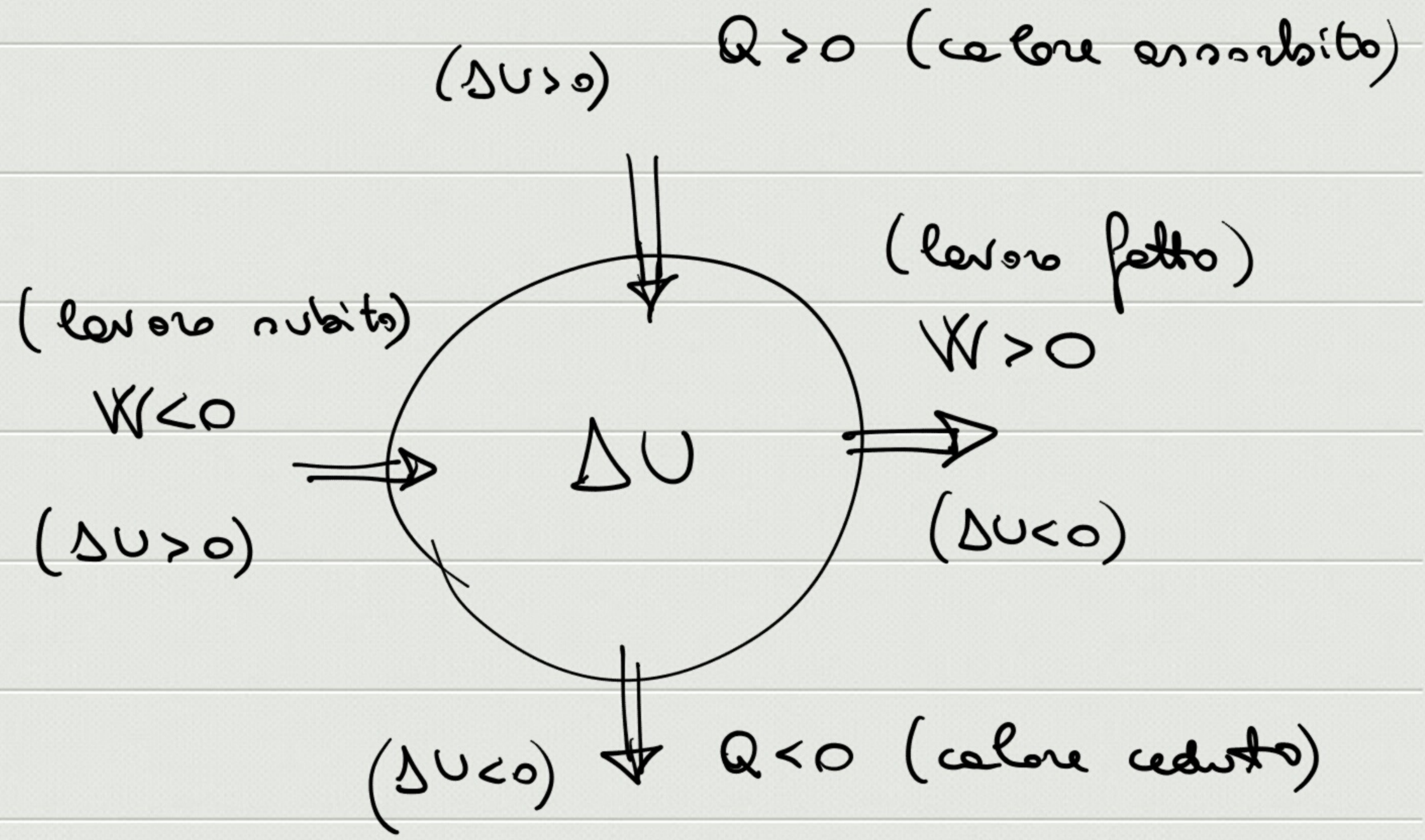
$$\boxed{\Delta U = Q_{pur.\ term.}}$$

W_{adiab} equivalente $Q_{pur.\ termico}$

$$\boxed{Q \sim W} \Rightarrow \boxed{Q = -W}$$

equivalenza calore - lavoro

$$\boxed{\Delta U = Q - W}$$



$$\boxed{Q = \Delta U + W}$$

1° principio della termodinamica

$U = U$ (coord. termodinamiche)

$U \rightarrow$ funzione di "stato"

$$\boxed{Q = \Delta U + W}$$: bilancio energetico

Q : calore è scambio di energia

$$Q_{i \rightarrow f} = \int_i^f \delta Q$$: dipende dalla trasformazione

Eq. 1° principio TD differenziale :

$$\boxed{\delta Q = dU + \delta W}$$